Cost Analysis

2026-06-22

Contents

1 Assumptions

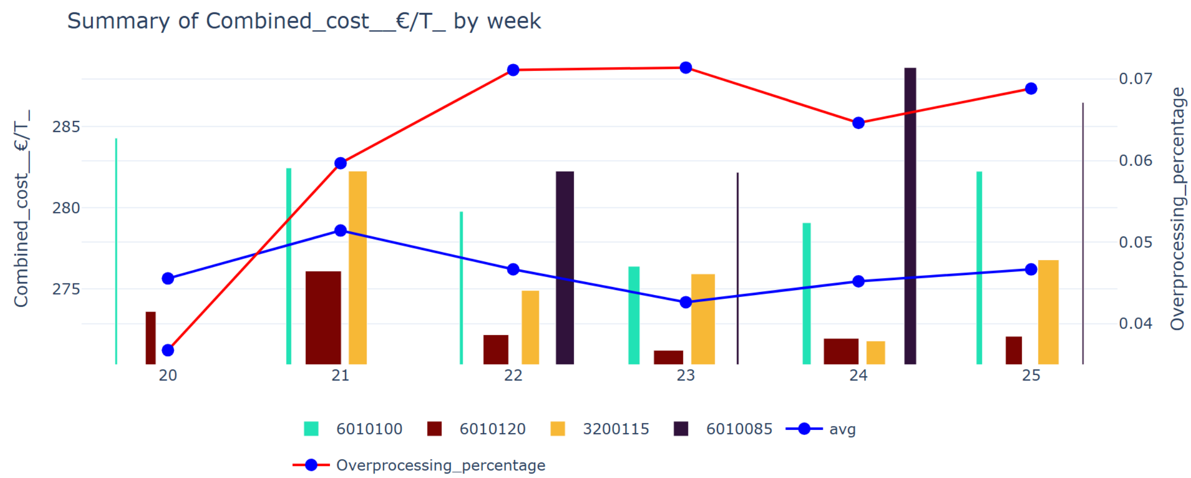
The aim of the analysis is to identify the most cost-effective production periods in months of data and recommend optimal operating windows for the corresponding key influencers that can be set as a benchmark for the weekly analysis.

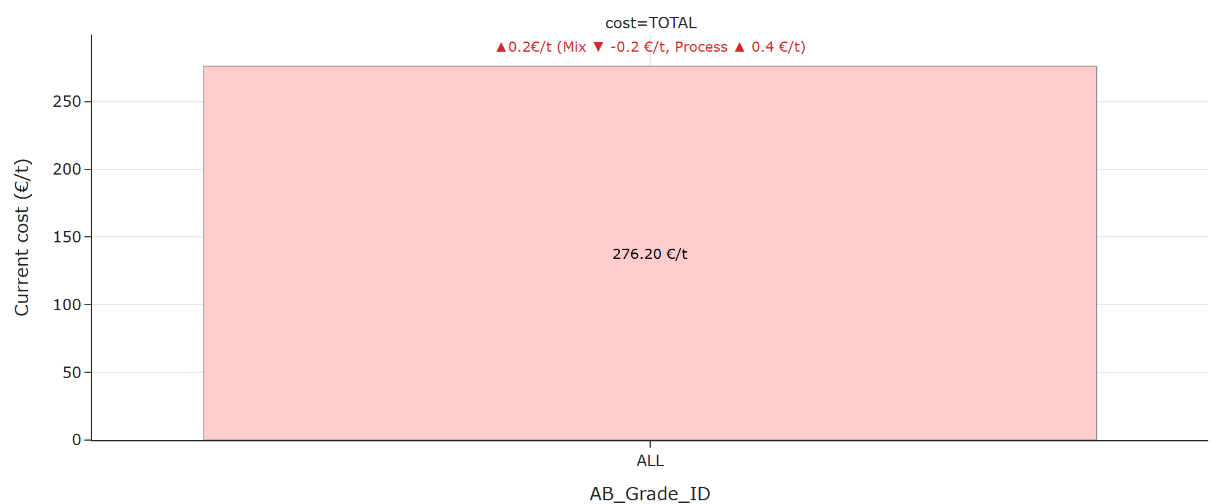
- **Period Analyzed**: May 17, 2026 – June 22, 2026
- **Cost Components Included**: Fibre, steam, electricity, starch, and chemicals.
- **Grades Included**: Only grades 6010085, 6010100, 601020 and 3200115 are included in this week's analysis.
- **Data Selection Criteria**: Only operating conditions where strength properties exceed specification thresholds were considered.
- **Naming Conventions Used**:
  - **Current**: Refers to the last week of the analyzed period.
  - **Historic**: Represents the average cost over the monthly interval previous to the "Current" period. Used as baseline.
  - **Best**: Corresponds to the cost associated with the optimal batch configuration.

2 Executive Summary

The following plot illustrates how total cost varies across different grades, enabling a comparison between the current configuration and the historical baseline.

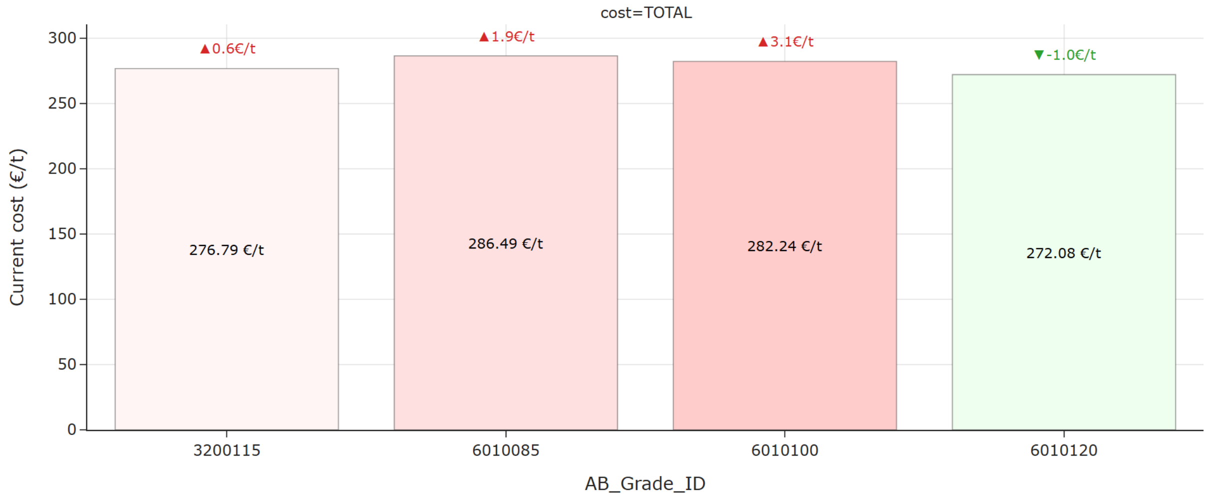
The next plot displays how total cost evolves over time, with a breakdown by grade to show individual contributions.





For all considered grades, TOTAL cost saw an increase of 0.20 €/t, resulting in 276.20 €/t.

Breaking down the overall cost change, -0.22 €/t was driven by a cheaper mix of grades, and +0.42 €/t was due to a deterioration in the process.



For grade 3200115, TOTAL cost saw an increase of 0.62 €/t, resulting in 276.79 €/t.

For grade 6010085, TOTAL cost saw an increase of 1.88 €/t, resulting in 286.49 €/t.

For grade 6010100, TOTAL cost saw an increase of 3.08 €/t, resulting in 282.24 €/t.

For grade 6010120, TOTAL cost saw a decrease of 0.97 €/t, resulting in 272.08 €/t.

Looking across individual grades, the strongest cost increase was observed for grade 6010100 (+3.08 €/t) and the strongest cost decrease was observed for grade 6010120 (-0.97 €/t). These movements stand out and may require further investigation.

Average change by component:

- **Starch__€/T** increased on average (+0.82 €/t).
- **Electricity__€/T** increased on average (+0.52 €/t).
- **Fibre_cost__€/T** decreased on average (-0.35 €/t).
- **Steam__€/T** increased on average (+0.19 €/t).
- **Chemicals__€/T** decreased on average (-0.03 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Starch__€/T** (+0.82 €/t on average), with the highest increase in grade 6010100 (+2.93 €/t) and the largest average decrease comes from **Fibre_cost__€/T** (-0.35 €/t on average), with the biggest drop in grade 6010085 (-1.50 €/t). These components stand out and may require further investigation.

Average change by steam:

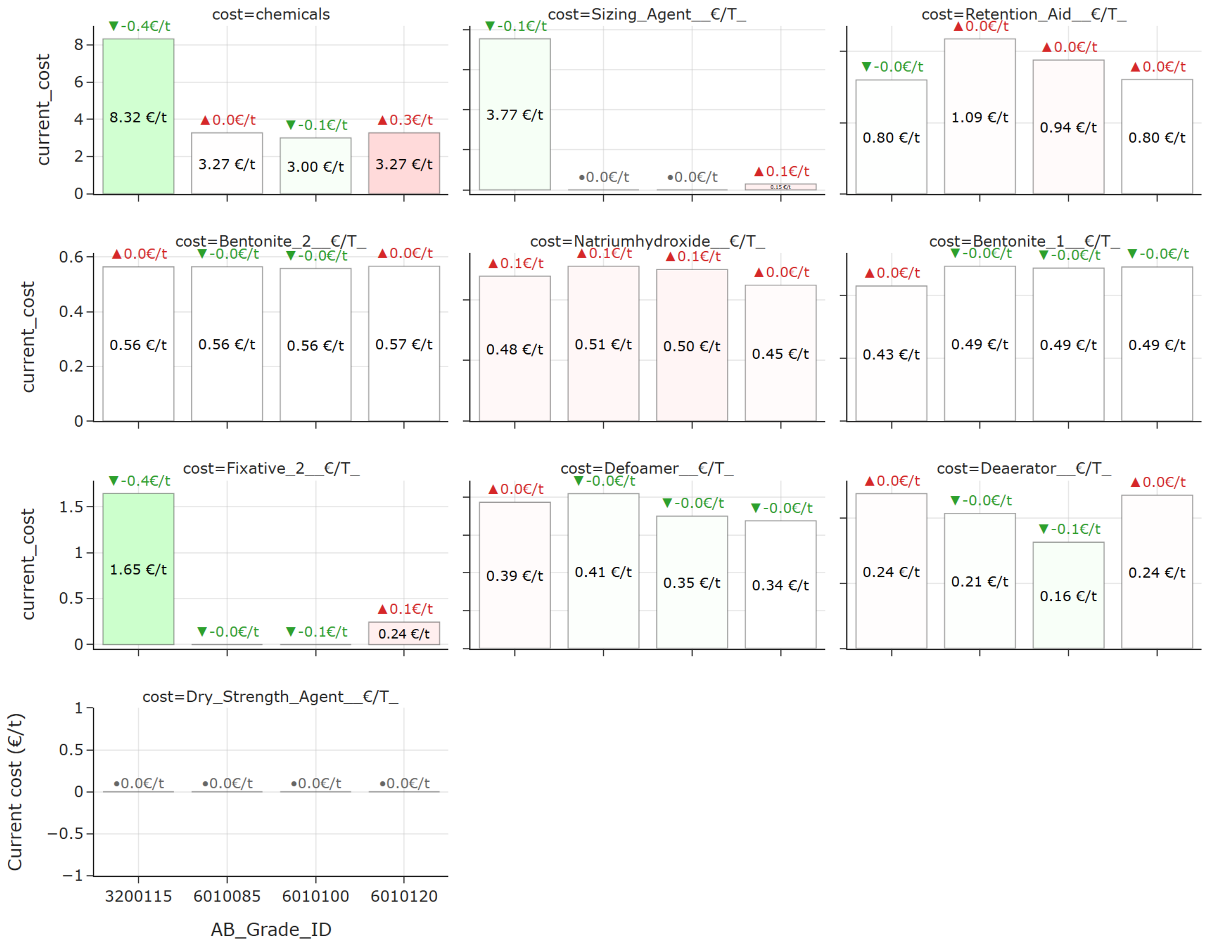
- **Steam_Predryers** decreased on average (-0.98 €/t).
- **Steam_Whitewater** increased on average (+0.59 €/t).
- **Steam_Afterdryers** increased on average (+0.50 €/t).
- **Steam_Heatexchangers** increased on average (+0.08 €/t).
- **Steam_Starchkitchen** increased on average (+0.01 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Steam_Whitewater** (+0.59 €/t on average), with the highest increase in grade 6010085 (+1.09 €/t) and the largest average decrease comes from **Steam_Predryers** (-0.98 €/t on average), with the biggest drop in grade 6010100 (-4.00 €/t). These components stand out and may require further investigation.

Average change by electricity:

- **Electricity_Other** increased on average (+0.55 €/t).
- **Electricity_Specific** increased on average (+0.15 €/t).
- **Electricity_Wiresection** decreased on average (-0.12 €/t).
- **Electricity_Afterdryer** decreased on average (-0.04 €/t).
- **Electricity_Press_and_Steambox** decreased on average (-0.03 €/t).
- **Electricity_Freqconverters** increased on average (+0.03 €/t).
- **Electricity_Predryer** decreased on average (-0.01 €/t).

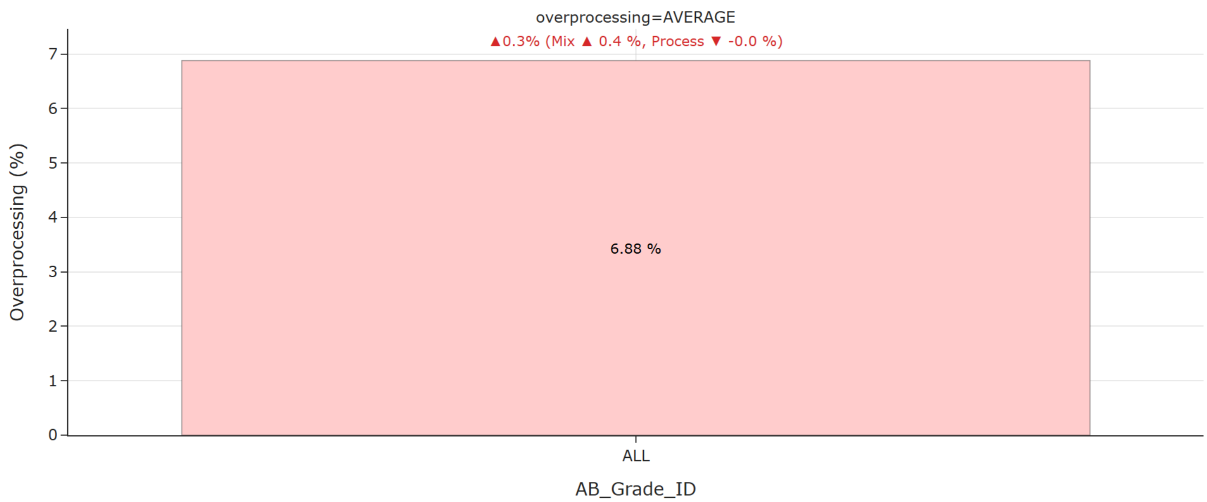
A more detailed breakdown by component shows that the largest average increase comes from **Electricity_Other** (+0.55 €/t on average), with the highest increase in grade 6010100 (+1.24 €/t) and the largest average decrease comes from **Electricity_Wiresection** (-0.12 €/t on average), with the biggest drop in grade 6010085 (-0.19 €/t). These components stand out and may require further investigation.



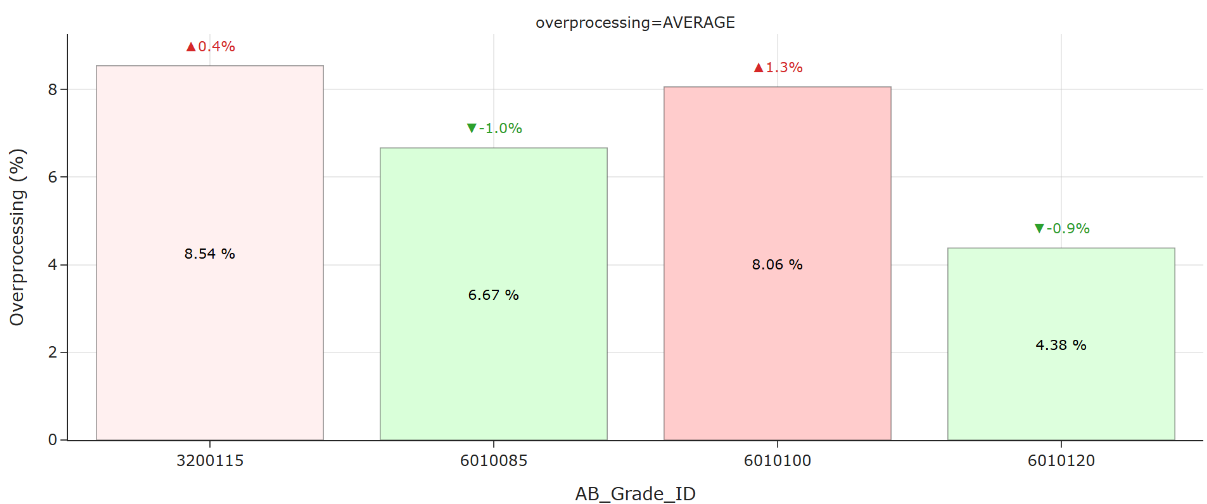
Average change by chemicals:

- **Fixative_2__€/T** decreased on average (-0.13 €/t).
- **Natriumhydroxide__€/T** increased on average (+0.06 €/t).
- **Sizing_Agent__€/T** increased on average (+0.02 €/t).
- **Retention_Aid__€/T** increased on average (+0.01 €/t).
- **Deaerator__€/T** decreased on average (-0.01 €/t).
- **Defoamer__€/T** decreased on average (-0.01 €/t).
- **Bentonite_1__€/T** increased on average (+0.00 €/t).
- **Bentonite_2__€/T** decreased on average (-0.00 €/t).

A more detailed breakdown by component shows that the largest average increase comes from ***Natriumhydroxide_€/T*** (+0.06 €/t on average), with the highest increase in grade 6010100 (+0.08 €/t) and the largest average decrease comes from ***Fixative_2_€/T*** (-0.13 €/t on average), with the biggest drop in grade 3200115 (-0.39 €/t). These components stand out and may require further investigation.



For all considered grades, TOTAL overprocessing saw an increase of 0.34 %, resulting in 6.88 %.



For grade 3200115, TOTAL overprocessing saw an increase of 0.38 %, resulting in 8.54 %.

For grade 6010085, TOTAL overprocessing saw a decrease of 1.00 %, resulting in 6.67 %.

For grade 6010100, TOTAL overprocessing saw an increase of 1.35 %, resulting in 8.06 %.

For grade 6010120, TOTAL overprocessing saw a decrease of 0.91 %, resulting in 4.38 %.

Looking across individual grades, the strongest overprocessing increase was observed for grade

6010100 (+1.35 %) and the strongest overprocessing decrease was observed for grade 6010085 (-1.00 %). These movements stand out and may require further investigation.

Average change by component:

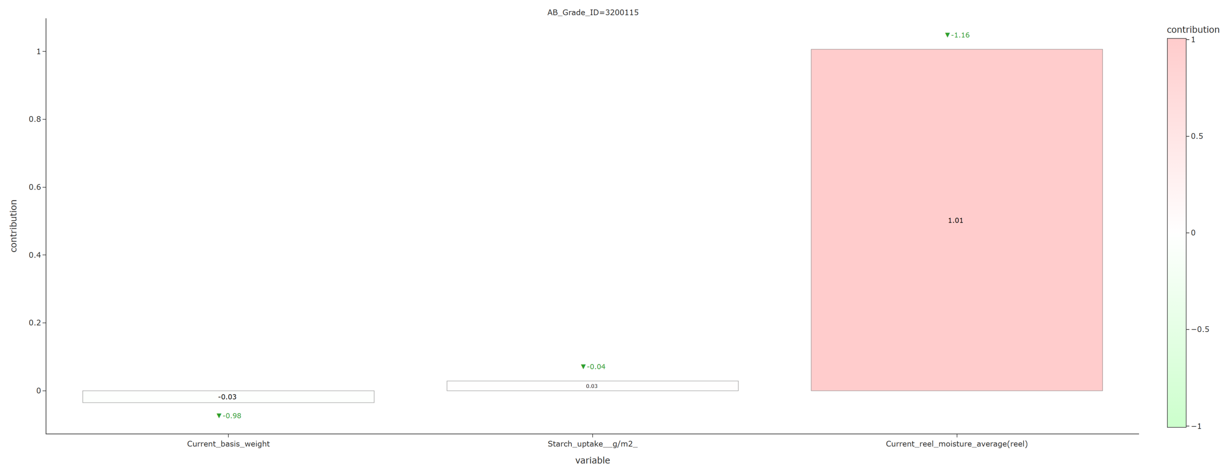
- **CMT30** decreased on average (-0.92 %).
- **Burst** increased on average (+0.92 %).
- **SCT_CD** increased on average (+0.40 %).

A more detailed breakdown by component shows that the largest average increase comes from **Burst** (+0.92 % on average), with the highest increase in grade 3200115 (+0.92 %) and the largest average decrease comes from **CMT30** (-0.92 % on average), with the biggest drop in grade 6010100 (-2.14 %). These components stand out and may require further investigation.

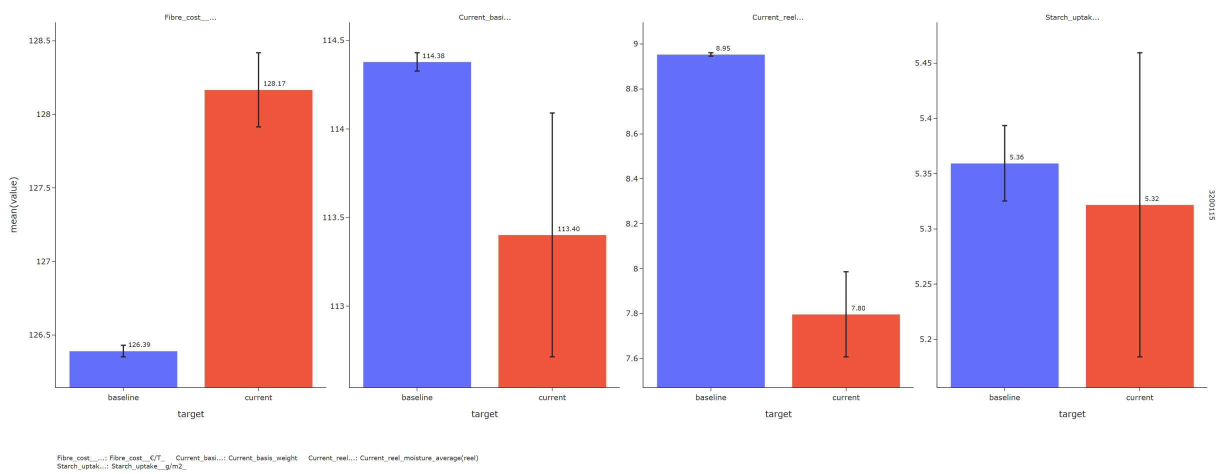
3 Breakdown by grades

3.1 3200115

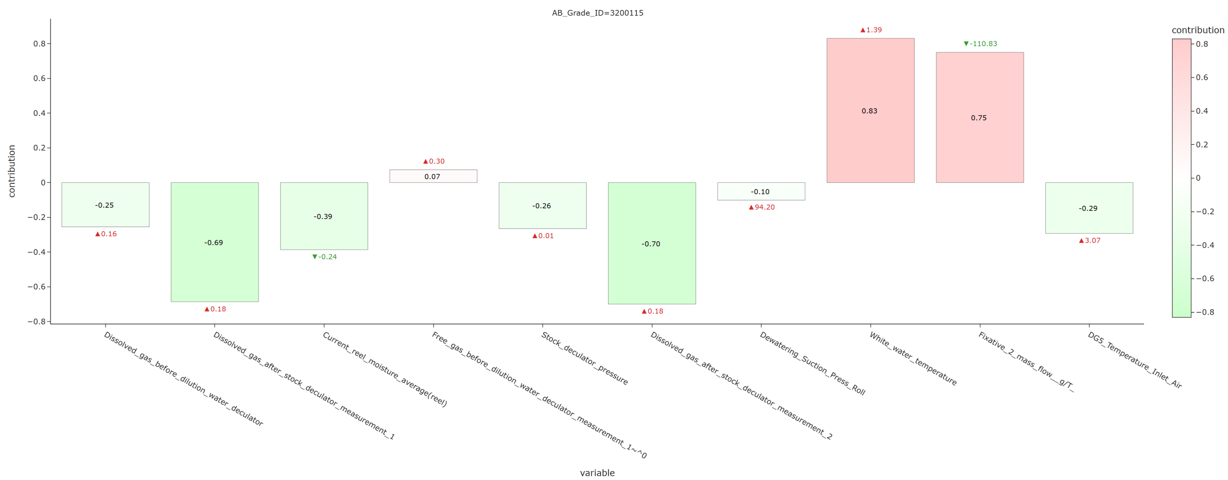
3.1.1 Fiber cost



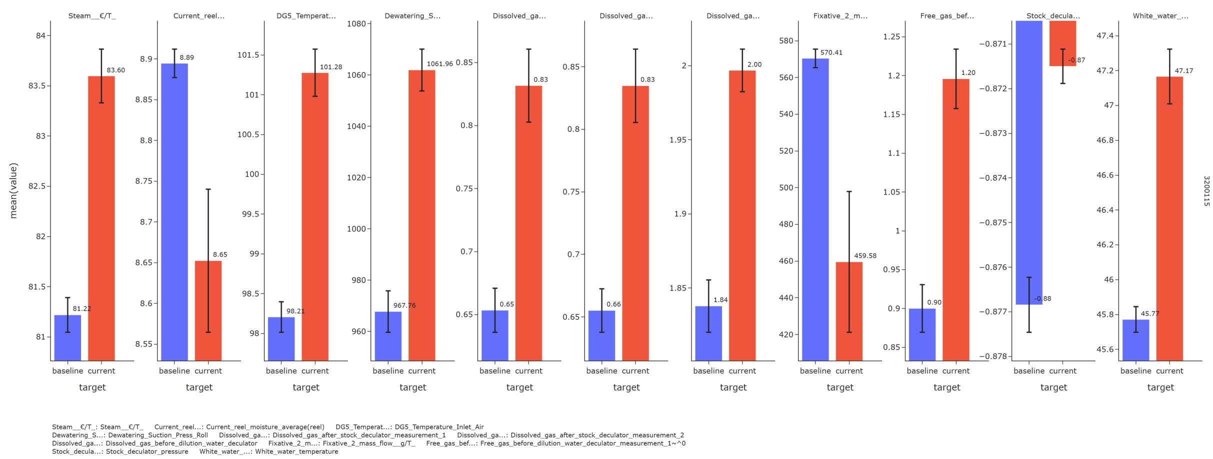
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-increasing drivers**: - Current reel moisture average(reel) decreased (-1.16)



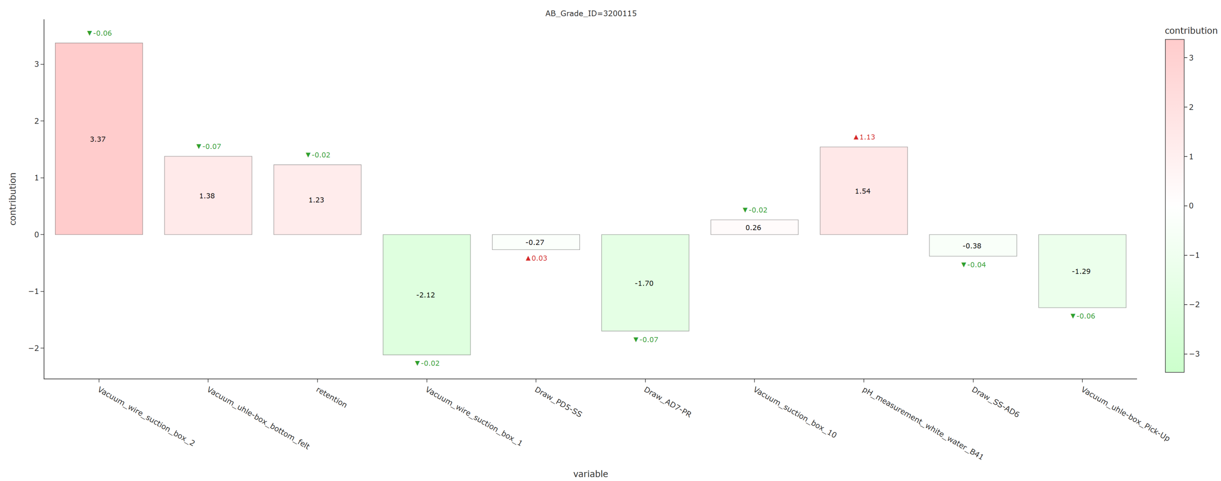
3.1.2 Steam cost



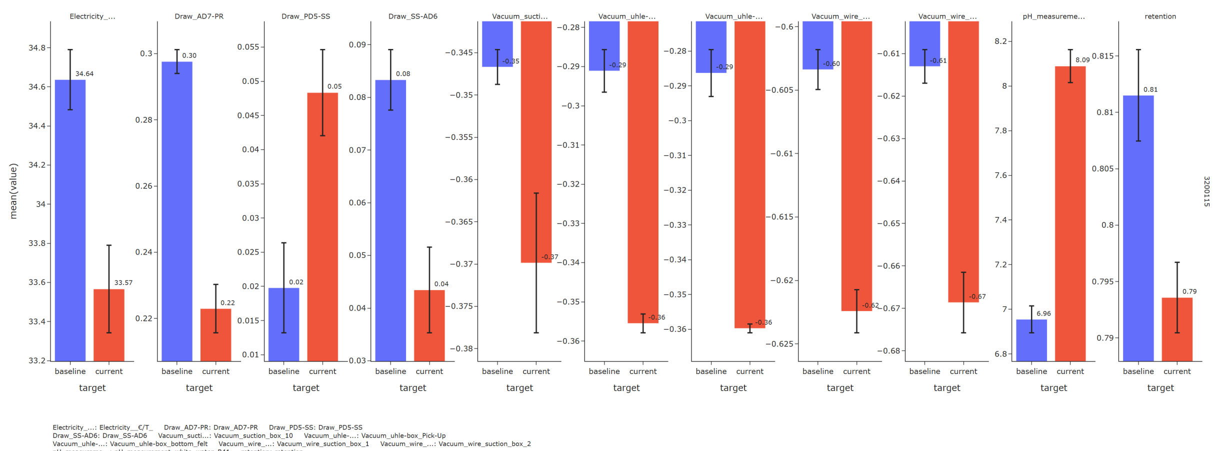
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Dissolved gas after stock deculator measurement 2 increased (+0.18) - Dissolved gas after stock deculator measurement 1 increased (+0.18) **Cost-increasing drivers**: - White water temperature increased (+1.39) - Fixative 2 mass flow g/T decreased (-110.83)



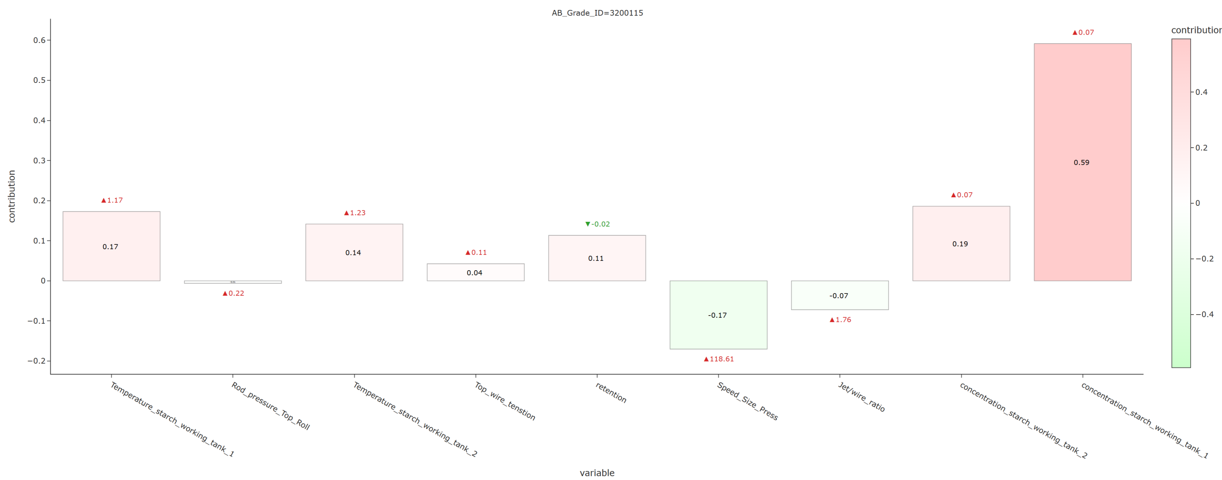
3.1.3 Electricity cost



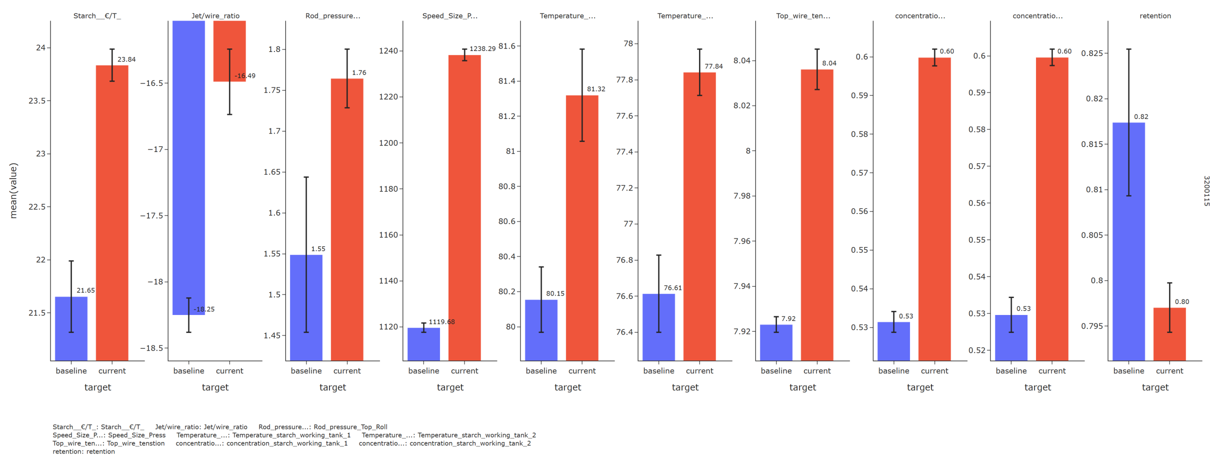
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Vacuum wire suction box 1 decreased (-0.02) - Draw AD7-PR decreased (-0.07) - Vacuum uhle-box Pick-Up decreased (-0.06) **Cost-increasing drivers**: - Vacuum wire suction box 2 decreased (-0.06) - PH measurement white water B41 increased (+1.13) - Vacuum uhle-box bottom felt decreased (-0.07)



3.1.4 Starch cost

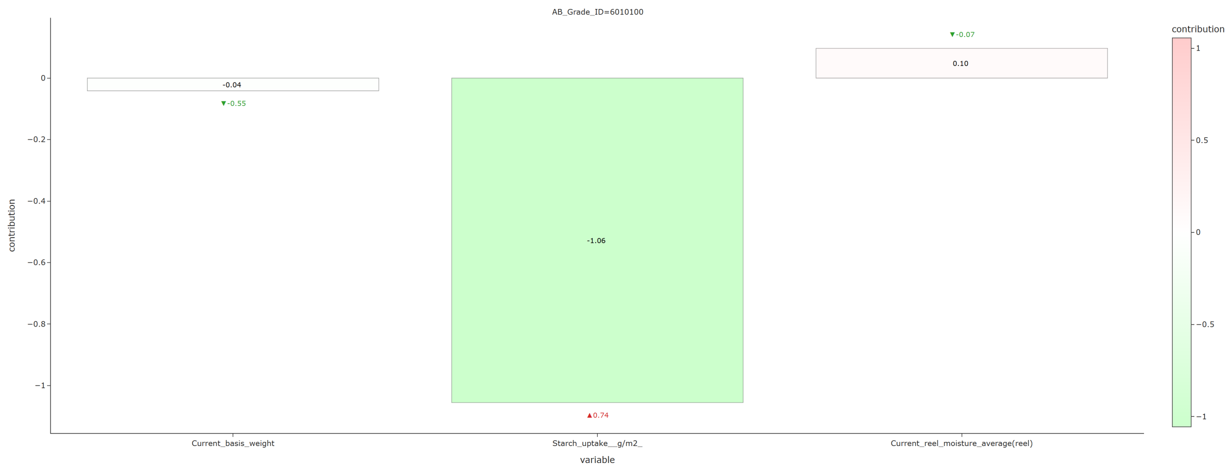


For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-increasing drivers**: - Concentration starch working tank 1 increased (+0.07) - Concentration starch working tank 2 increased (+0.07)

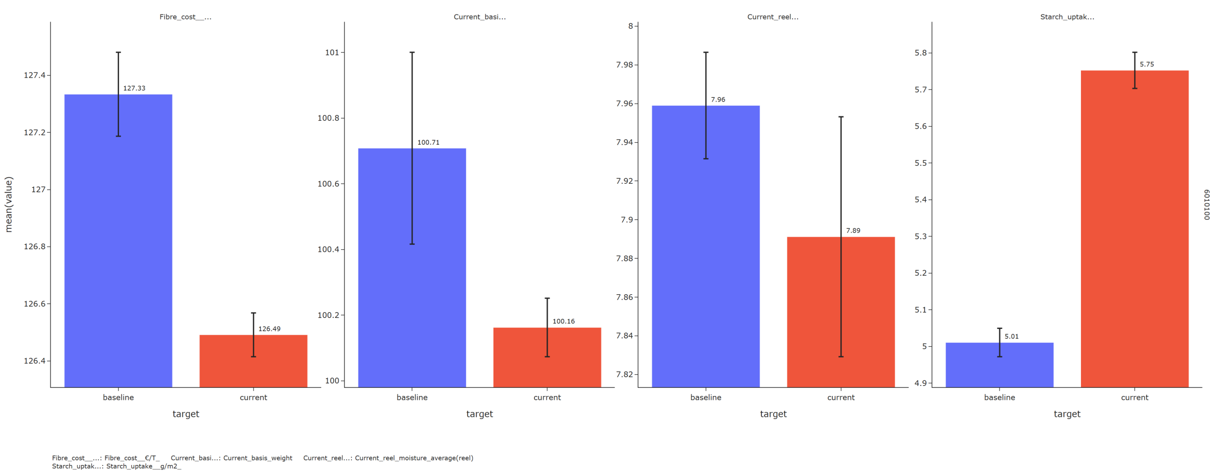


3.2 6010100

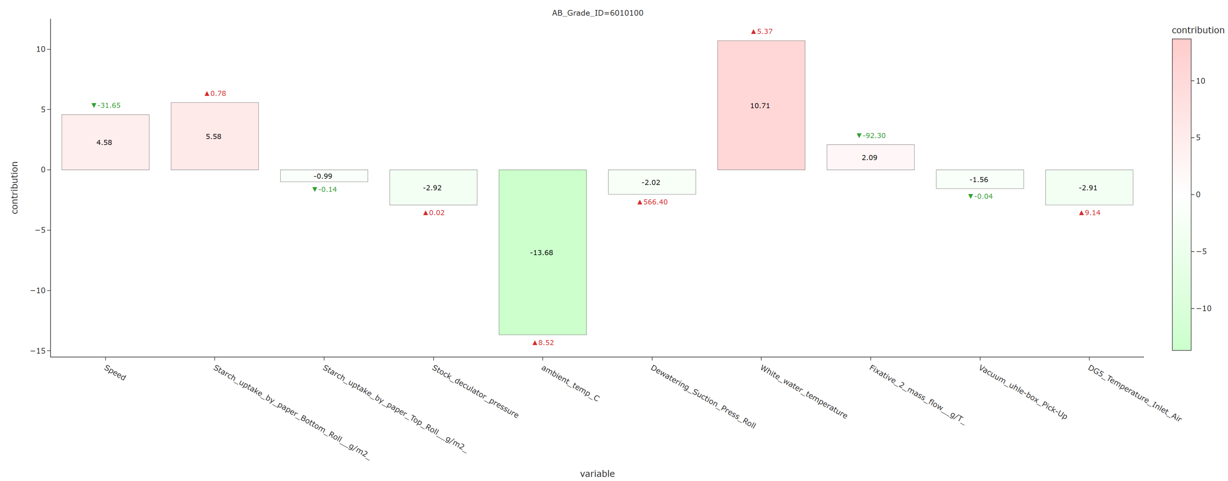
3.2.1 Fiber cost



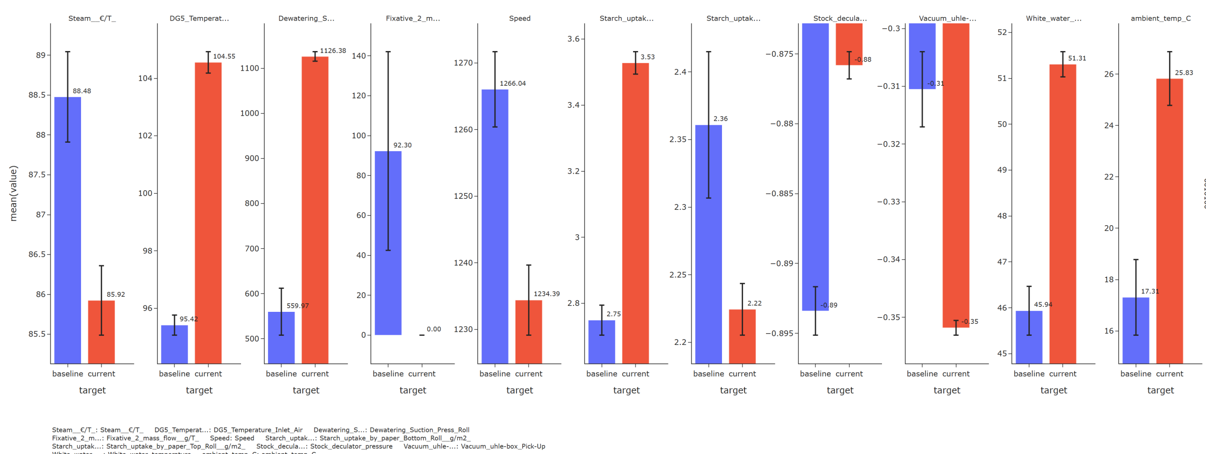
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Starch uptake g/m2 increased (+0.74)



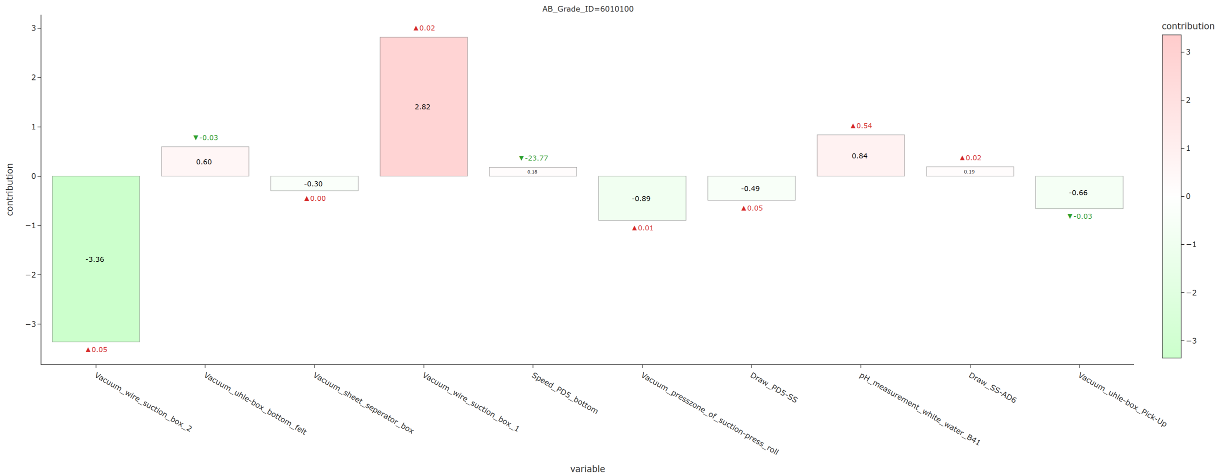
3.2.2 Steam cost



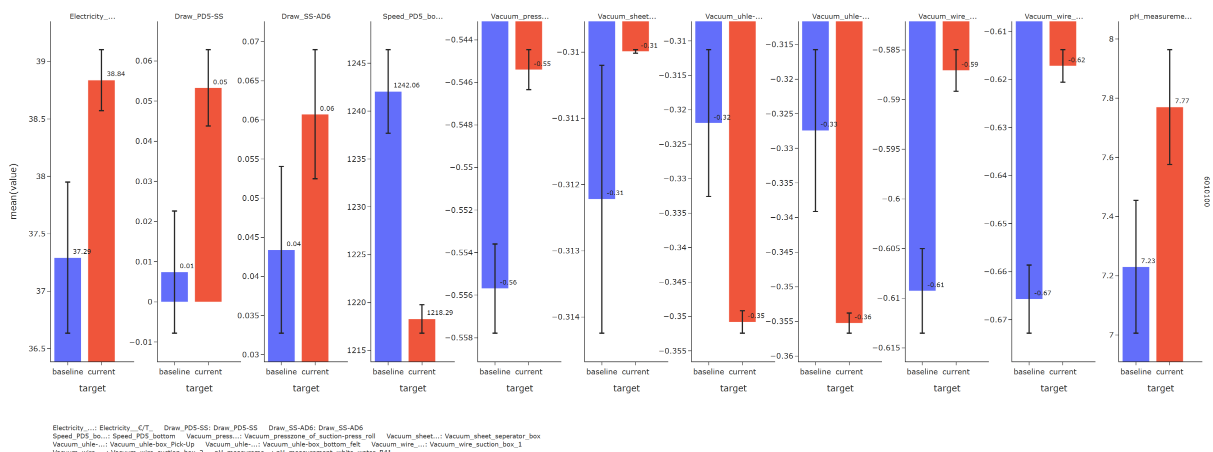
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Ambient temp C increased (+8.52) - Stock deculator pressure increased (+0.02) **Cost-increasing drivers**: - White water temperature increased (+5.37) - Starch uptake by paper Bottom Roll g/m2 increased (+0.78) - Speed decreased (-31.65)



3.2.3 Electricity cost

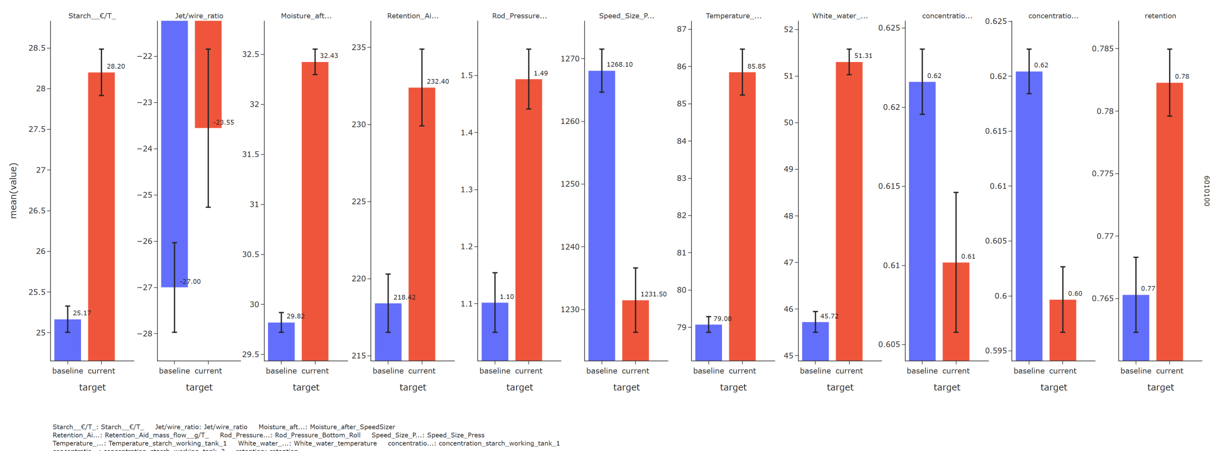


For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Vacuum wire suction box 2 increased (+0.05) - Vacuum presszone of suction-press roll increased (+0.01) - Vacuum uhle-box Pick-Up decreased (-0.03) **Cost-increasing drivers**: - Vacuum wire suction box 1 increased (+0.02) - PH measurement white water B41 increased (+0.54) - Vacuum uhle-box bottom felt decreased (-0.03)



3.2.4 Starch cost

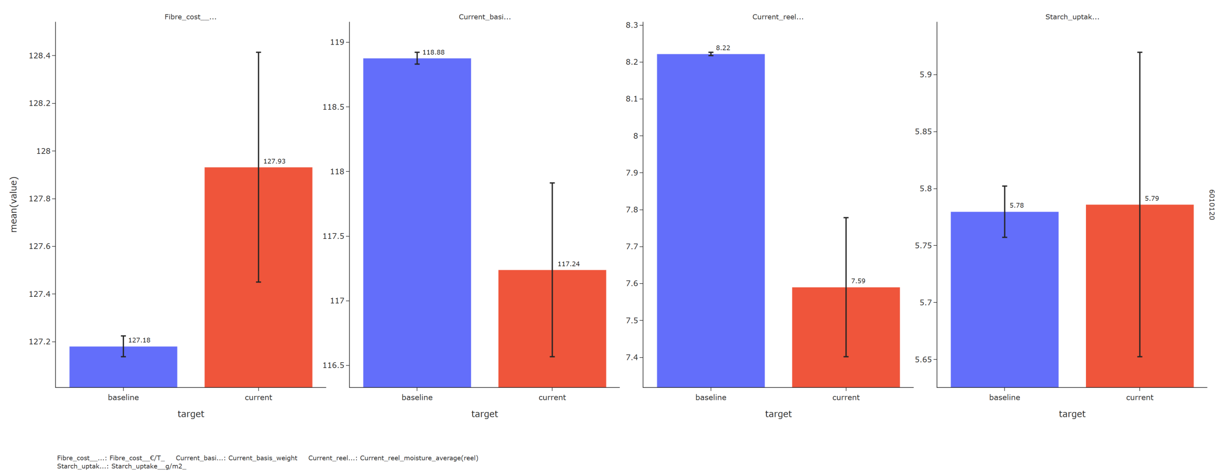
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Jet/wire ratio increased (+3.45) **Cost-increasing drivers**: - Temperature starch working tank 1 increased (+6.77) - Moisture after SpeedSizer increased (+2.61)



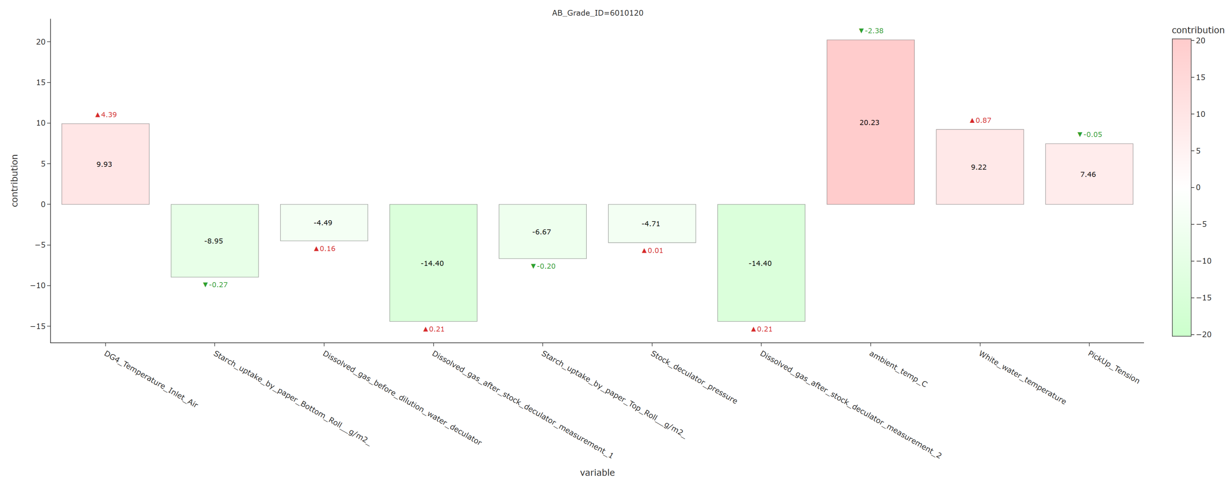
3.3 6010120

3.3.1 Fiber cost

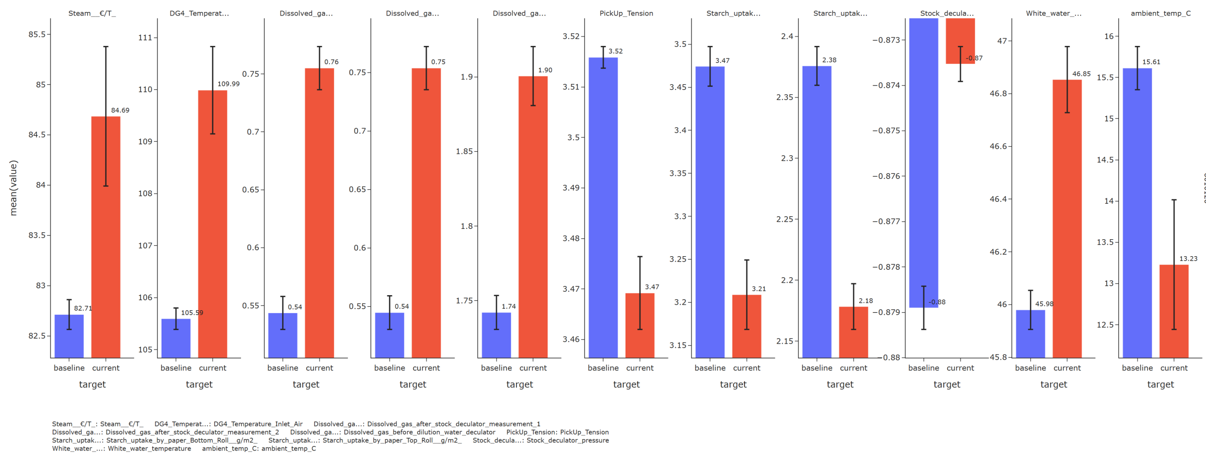
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-increasing drivers**: - Current reel moisture average(reel) decreased (-0.63)



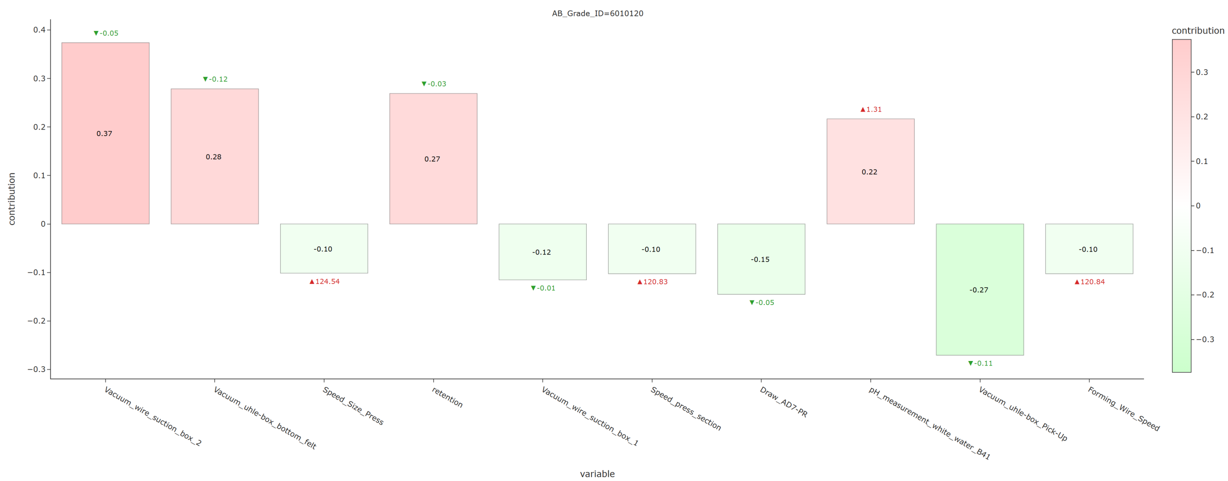
3.3.2 Steam cost



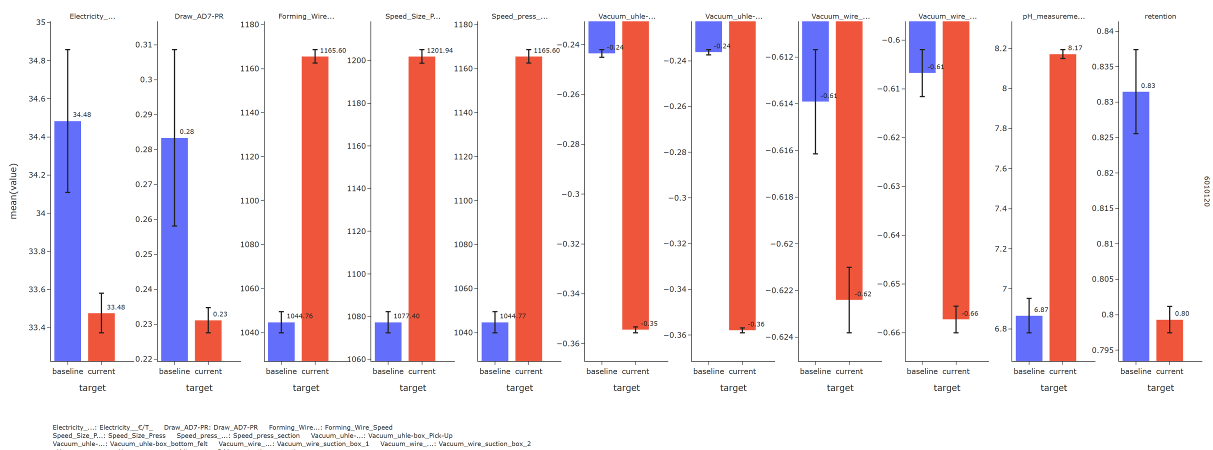
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Dissolved gas after stock deculator measurement 1 increased (+0.21) - Dissolved gas after stock deculator measurement 2 increased (+0.21) **Cost-increasing drivers**: - Ambient temp C decreased (-2.38) - DG4 Temperature Inlet Air increased (+4.39)



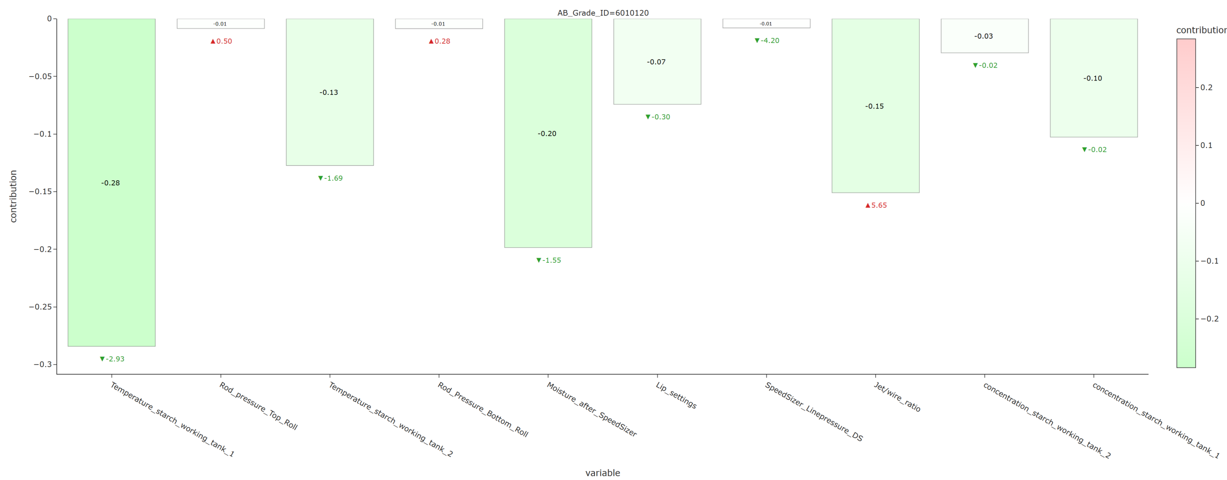
3.3.3 Electricity cost



For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Vacuum uhle-box Pick-Up decreased (-0.11) - Draw AD7-PR decreased (-0.05) **Cost-increasing drivers**: - Vacuum wire suction box 2 decreased (-0.05) - Vacuum uhle-box bottom felt decreased (-0.12) - Retention decreased (-0.03) - PH measurement white water B41 increased (+1.31)



3.3.4 Starch cost



For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Temperature starch working tank 1 decreased (-2.93) - Moisture after SpeedSizer decreased (-1.55) - Jet/wire ratio increased (+5.65)